# Faraday Excitation: Math512 Milestone 5

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## 1 Introduction

In 1831, Michael Faraday published a paper describing an observation of an unusual phenomenon [1]. He observed that when different liquids in a variety of containers underwent vertical oscillations of various frequencies, the superposition of surface waves produced various surface patterns of standing waves. This phenomenon came to be known as Faraday Excitation.

Faraday's report was written in a classical style from the early 19th century in the sense that it was more qualitative in its description rather that quantitative. Since there was a lack of a theoretical explanation, many researchers over the years have conducted different experiments in an effort to find a quantitative description for various systems. One problem with this is that there are many different forms of liquids, containers and frequencies to be used and thus there will be a copious amount of theoretical models that only work for specific variables. Therefore, different equations will be needed for different systems.

This report describes the research and investigation of Faraday Excitation of tap water in a cylindrical container. This report could lead to how to minimize surface waves by optimizing the shape of the container. By examining the effects of fluid depth, frequency of the vertical oscillator and shape of the container, it should be possible to determine the best shape to minimize the sloshing of fluid or undesirable wave patterns in cylindrical containers due to vertical oscillations.

These experiments use a mechanical vertical oscillator with a cylindrical container mounted on a vertical oscillator. Water is chosen for the fluid because it constitutes a realistic scenario and the fact that it is readily available. The vertical oscillator is powered by an electronic wave generator with an amplifier. These are used to produce both the amplitude and the type of wave (with dimensions) that will oscillate the container in

the vertical direction. A high-speed camera is setup for recording so that the team can observe the otherwise impossible maximums and minimums of the surface pattern.

In investigating previous work on Faraday Excitation, it was found that there has been much related work done on the topic. It seemed that for every combination of shape of container and fluid type, there was another paper written.

Some of the prior research that was done was found to be quite complicated. Forrest [2] and Merkt [3] conducted experiments using non-newtonian fluids and observed that the complex behavior of a non-Newtonian fluid makes the mathematical modeling quite complicating. For example, Merkt used corn starch and throughout time, it became so thick that the maximums and minimums didn't oscillate but grew in amplitude. The maximums produced blobs growing out from the edge of the holes that were where the minimums were. Their results helped further persuade our team to use a simple Newtonian fluid such as water in order to keep the calculations as simple as possible. Mandal used a thin film to dampen the oscillations in his experiments [4]. This is basically putting another liquid that is less dense onto the surface of a fluid and the result is that it should form a coating that will dampen out the surface pattern to a proportional degree. Again, in keeping the experiment as simple as possible, we avoided the use of multiple fluids.

What was needed was some insight on how to understand our experiment mathematically. Later, our team came across some more appealing writings. Yong-Jun used vertical oscillations on a cylindrical container to model surface wave patterns, however this research wasn't very informative in its quantitative expressions [5]. The formulas were merely presented rather than being derived. Therefore, we resorted to several books on surface wave patterns for better explanations. Lighthill wrote one fairly comprehensive book on the topic of water waves that breaks down the math to much simpler equations with more detailed descriptions (section 2.2 and 2.3) [6]. He provides an in-depth analysis of the boundary of a fluid surface. Since the surface of a fluid in a cylindrical container can be modeled by incorporating the use of Bessel functions, Haberman introduces the basics that are found in section 2.4 [7].

Combining these equations our team was able to produce computer models (via MAPLE) and run simulations that could later be compared/tested with actual data from real laboratory simulations.

# 2 Modeling

The goal is to develop a model that accurately predicts the formation of surface waves due to the vertical excitations of the fluid. Since the cylindrical container has been mentioned in the literature reviews and a few of ideal dimensions (i.e. big enough to keep the liquid from splashing out) have been obtained quite readily, it was determined to focus on using cylindrical containers.

The surface waves patterns on a vertically oscillating are the superposition of waves. Therefore, our model will require the use of Bessel functions since they are a function of the separation of variables. These Bessel equations will then be combined with the mechanics of a fluid surface in order to produce a workable model.

## 2.1 Assumptions

To arrive at the model for surface waves on a cylinder, the assumptions made are as follows:

- The container is a perfect cylinder with uniform sides and bottom. In other words, it is assumed that the sides are completely vertical and the deformations in the sides and bottom have a negligible effect on the formation of the surface waves.
- The fluid is not excited to the point where droplets are emitted from the fluid surface.
- Assume the fluid can be described by the Navier-Stokes equation in the case of laminar flow and constant viscosity (In reality, this assumption can be relaxed, since it only affects equation (1), but it makes the linear terms more apparent).
- Assume the container has a cylindrical geometry with radius, $R$ and height, $H$, described by a cylindrical coordinate system $(r, \theta, z)$ with origin at the center of the bottom surface of the container.

## 2.2 Developing the Model

To model the evolution of the surface waves over time, start with the Navier-Stokes equation [8] for an incompressible fluid undergoing laminar flow, which describes the conservation of momentum within the fluid:

$$\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = \rho\mathbf{g} - \nabla P + \mu\nabla^2\mathbf{u} \tag{1}$$

In the Navier-Stokes equation:

- $\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t}$ is the material (also called the convective, or Stokes) derivative of the velocity, $\mathbf{u}$ with respect to time, $t$
- $\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = \frac{\partial\mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}$
- $\rho$ is the density of the fluid
- $\mathbf{g}$ is the gravity vector
- $P$ is the pressure of the fluid and
- $\mu$ is the static viscosity of the fluid

Linearizing equation (1) yields:

$$\rho\frac{\partial\mathbf{u}}{\partial t} = \rho\mathbf{g} - \nabla P \tag{2}$$

Define the hydrostatic pressure, $P_0$, and excess pressure, $P_e$, as follows,

$$P_0 = P_{atm} - \rho g z \tag{3}$$
$$P_e = P - P_0 \tag{4}$$

where $P_{atm}$ is atmospheric pressure.

Using those definitions, equation (2) can be rewritten as:

$$\rho\frac{\partial\mathbf{u}}{\partial t} = -\nabla P_e \tag{5}$$

One can then take the curl of both sides as follows:

$$\rho\left(\nabla\times\frac{\partial\mathbf{u}}{\partial t}\right) = \nabla\times(-\nabla P_e) \Rightarrow$$
$$\rho\frac{\partial(\nabla\times\mathbf{u})}{\partial t} = \mathbf{0} \tag{6}$$

Following from the identity that the curl of the gradient of a scalar function is zero, equation (6) states that the vorticity of the fluid, $\nabla\times\mathbf{u}$ does not vary with time. The vorticity of the fluid represents the rotational component of the velocity, and since it

doesn't vary with time, it does not affect the pressure distribution of the fluid. Thus, we will neglect it and assume that the field is irrotational, since the irrotational component of the velocity field *does* change with time. An irrotational velocity field can be represented with a velocity potential, $\psi$, where $\nabla\psi = \mathbf{u}$.

Conservation of mass must also be fulfilled, and can be described by the following equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{7}$$

Since the fluid is incompressible, its density does not vary with time, and thus equation (7) simplifies to:

$$\begin{aligned} \nabla \cdot \mathbf{u} = 0 &\Rightarrow \\ \nabla \cdot \nabla\psi = 0 &\Rightarrow \\ \nabla^2\psi &= 0 \end{aligned} \tag{8}$$

Thus, the velocity field of the fluid can be found via the Laplace equation shown in equation (8).

## 2.3 Boundary Conditions

Let the height of the fluid in the container be defined by $\zeta = \zeta(r, \theta, t)$. At the surface of the fluid, $z = \zeta$. Assuming a free surface with no surface tension, the pressure of the fluid at the air-fluid interface equals atmospheric pressure. From equations (3) and (4), this implies that:

$$P_e = \rho g \zeta \tag{9}$$

Returning to equation (2) and substituting $\nabla\psi = \mathbf{u}$, then integrating both sides yields:

$$P_e = -\rho \frac{\partial \psi}{\partial t} \tag{10}$$

It can then be seen from equations (9) and (10) that:

$$\frac{\partial \psi}{\partial t} = -g\zeta \tag{11}$$

This will be useful shortly. By definition, it is known that the acceleration of the surface should equal the change in the velocity at the surface:

$$\frac{\partial \zeta}{\partial t} = \left(\frac{\partial \psi}{\partial z}\right)|_{z=\zeta} \tag{12}$$

Returning to equation (11), apply a time derivative to both sides:

$$\frac{\partial^2 \psi}{\partial t^2} = -g \frac{\partial \zeta}{\partial t} \tag{13}$$

One can then substitute the result of equation (12) into (13) to get a boundary condition for the top surface of the fluid:

$$\frac{\partial^2 \psi}{\partial t^2} = -g \left( \frac{\partial \psi}{\partial z} \right) |_{z=\zeta} \tag{14}$$

The remaining boundary conditions are as follows:

- $\frac{\partial \psi}{\partial z}(r, \theta, 0, t) = 0$, since no fluid flows through the bottom of the container.
- $\frac{\partial \psi}{\partial r}(R, \theta, z, t) = 0$, since no fluid flows through the sides of the container.
- $\psi(0, \theta, z, t)$ is bounded

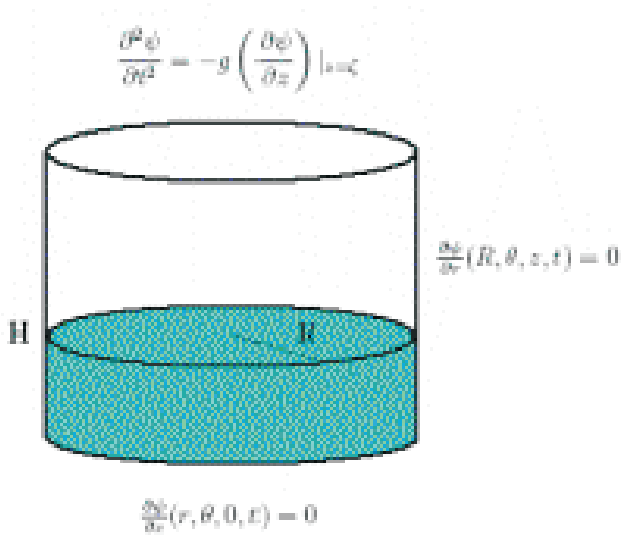


Figure 1: Representation of a cylindrical container with boundary conditions for the sides, bottom and top. Notice that the sides and bottom equal zero, this represents the fact that fluid can't flow past the container walls. Also, H represents the height of water and R is the radius of the container (with straight sides).

## 2.4 Solving the Laplacian

Let $\omega$ be a sine/cosine wave oscillation frequency. Define nondimensional quantities as follows:

- Define nondimensional time, $\bar{t} = \omega t$.
- Define nondimensional radius, $\bar{r} = \frac{r}{R}$.
- Define nondimensional vertical coordinate, $\bar{z} = \frac{z}{H}$.
- Define nondimensional velocity potential, $\bar{\psi} = \frac{\psi}{R^2\omega}$.

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Changing variables in equation (8), the nondimensional model has almost exactly the same form:

$$\nabla^2\bar{\psi} = 0 \tag{15}$$

Since the oscillations are sine/cosines, assume:

$$\bar{\psi}(\bar{r}, \theta, \bar{z}, \bar{t}) = \exp(i\bar{t})\phi(\bar{r}, \theta, \bar{z}) \tag{16}$$

Substituting equation (16) into equation (15) yields:

$$\nabla^2\phi = 0 \tag{17}$$

In equation (17) and equation (8), let $\phi = f(\bar{r})G(\theta)h(\bar{z})$ and solve by separation of variables. Substituting in for $\phi$ gives

$$f''Gh + \frac{1}{\bar{r}}f'Gh + \frac{1}{\bar{r}^2}fG''h + fGh'' = 0 \tag{18}$$

$$G(0) = G(2\pi) \tag{19}$$

$$G'(0) = G'(2\pi) \tag{20}$$

$$f(0) \text{ is bounded} \tag{21}$$

$$f'(1) = 0 \tag{22}$$

$$h'(0) = 0 \tag{23}$$

$$h(\frac{H}{R}) = \frac{-gh'(\frac{H}{R})}{R^2\omega} \text{ from equation (14)} \tag{24}$$

Divide through equation (18) by $fGh$. This gives

$$\frac{f''}{f} + \frac{1}{\bar{r}}\frac{f'}{f} + \frac{1}{\bar{r}^2}\frac{G''}{G} + \frac{h''}{h} = 0 \tag{25}$$

Rearranging to isolate $G$ gives:

$$\bar{r}^2 \frac{f''}{f} + \bar{r}\frac{f'}{f} + \bar{r}^2 \frac{h''}{h} = -\frac{G''}{G} \tag{26}$$

Next, solve

$$\frac{G''}{G} = -\mu^2 \tag{27}$$

where $\mu$ is real, anticipating that the boundary conditions ($G(0) = G(2\pi)$ and $G'(0) = G'(2\pi)$) will result in periodicity.

Solving (27) gives the solution

$$g(\theta) = C_1 \sin(\mu\theta) + C_2 \cos(\mu\theta) \tag{28}$$

$\mu = n$ where $n$ is a positive integer. Thus, the solutions become

$$g(\theta) = C_1 \sin(n\theta) + C_2 \cos(n\theta) \tag{29}$$

where $C_1$ and $C_2$ are constants.

Returning to equation (26), substitute using equation (27) and rearrange to isolate $h$:

$$\frac{f''}{f} + \frac{f'}{f}\frac{1}{\bar{r}} - \frac{\mu^2}{\bar{r}^2} = -\frac{h''}{h} \tag{30}$$

Then, set the right hand side equal to a constant, $-\kappa^2$, where $\kappa$ is also real. Then equation (30) becomes a Bessel equation:

$$\bar{r}^2 f'' + \bar{r} f' + (\kappa^2 \bar{r}^2 - \mu^2) f = 0 \tag{31}$$

Solving for $f$:

$$f(\bar{r}) = \begin{cases} C_3 J_\mu(\kappa\bar{r}) + C_4 Y_\mu(\kappa\bar{r}), & \kappa \neq 0 \\ C_5 \bar{r}^n + C_6 \bar{r}^{-n}, & \kappa = 0 \end{cases} \tag{32}$$

The boundedness condition in equation (21) implies that $C_4$ and $C_6$ equal zero. Applying the "no flux" boundary condition in equation (22) implies that $C_5$ equals zero (thus $\kappa$ cannot equal zero) and that

$$\begin{aligned} J_n'(\kappa) &= 0 \\ \kappa &= z_{mn} \end{aligned} \tag{33}$$

where $z_{mn}$ is the $m$th zero of the derivative of the Bessel function of order $n$. Thus, the solution to the $f$ problem is:

$$f(\bar{r}) = C_3 J_\mu(z_{mn}\bar{r}) \tag{34}$$

Returning to equation (30), this time, set the left hand side equal to $-\kappa^2$.

$$\frac{h''}{h} = \kappa^2 \tag{35}$$

Rearranging and solving, it is found that:

$$h(\bar{z}) = C_7 \cosh(\kappa\bar{z}) + C_8 \sinh(z_{mn}\bar{z}) \tag{36}$$

Applying the bottom "no flux" boundary condition in equation (23) implies that $C_7$ equals zero. Applying the top "free surface" boundary condition implies that $C_8 = g \tanh\left(\frac{H\kappa}{R}\right)$, or

$$h(\bar{z}) = \frac{g}{R^2\omega} \tanh\left(\frac{H\kappa}{R}\right) \sinh(z_{mn}\bar{z}) \tag{37}$$

Returning to a general solution for $\phi$, combine equations (29), (34), and (36):

$$\phi(\bar{r},\theta,\bar{z}) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \left(A_{mn} \sin(n\theta) J_n(z_{mn}\bar{r}) \sinh(z_{mn}\bar{z}) + B_{mn} \cos(n\theta) J_n(z_{mn}\bar{r}) \cosh(z_{mn}\bar{z})\right) \tag{38}$$

where $A_{mn}$ and $B_{mn}$ are constants to be found later.

# 3 Experimental Design

For our cylindrical container, we initially used the top of a CD holder because of its straight vertical sides, tall height and transparency. There were notches in the bottom so in order to make the bottom uniformly flat, we covered the holes with a silicon sealant. The container was then tested on the vertical oscillator with 5 cm of water inside and with the use of an accelerometer and a computer package called Data Studio, it was found not to be too heavy for the oscillator.

Familiarization with the system allowed us to become more comfortable with the equipment such as the amplifier, wave generator and Data Studio. Frequencies between 2.5

and 100 hz were found to be ideal for producing visual surface wave patterns. Anything above 100 hz and the peaks became so close together that even the use of a strobe light couldn't show the pattern. Also our input amplitudes had to decrease as frequency increased because splashing began to occur and that was against our initial assumptions. We observed that different wave patterns, such as square, sine and triangle produced different surface patterns when the frequencies and amplitudes were kept constant. Also, on low frequencies the amplitude had a limit on how high it could go. In addition to using very low frequencies ( below 15hz) and in sine wave mode, the peaks would flatten out when the gain was increased. These equipment limitations will have to be kept in mind for future experiments.

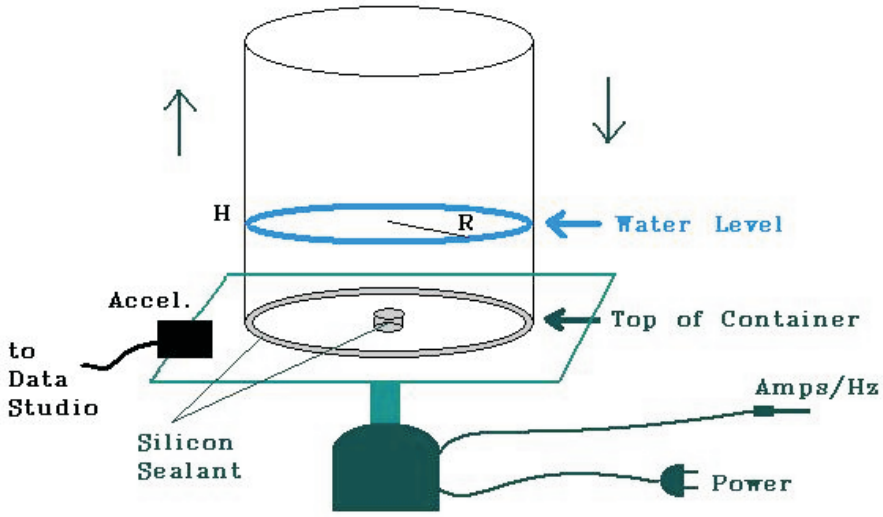


Figure 2: This is a sketch of our general design in the initial stages. Due to some of the noise produced by the design, the table was changed to a circular one and the accelerometer was removed. The container was also replaced with one of a smaller height. Initial conditions; Radius = 6.24cm, Water height = 5 cm

# 4 Computer Simulations

Before any experiments were conducted our computer simulations need to be run. Using a computer package called MAPLE, we inserted our equations and let the computer not only solve them but plot them for us in 3-D. This should be appreciated for the fact that it took a quite a bit of time to develop and comprehend. So much time that we moved into the laboratory to collect data in order to keep from falling to far behind schedule. Final comprehension of the simulations and their comparisons to the laboratory results would come later and be a minor set back.

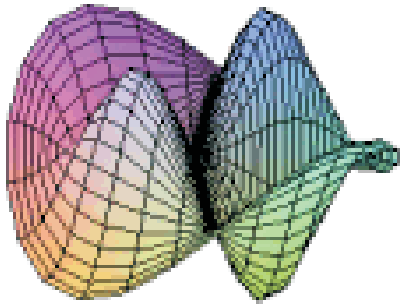

Figure 3: This is a MAPLE computer simulation of a 6.24 radius cylindrical container with a water height of 0.7cm and modes of m=2, n=2.

# 5 Equipment Noise

Due to time constraints, we proceeded into the laboratory before the computer simulations were completed so that we could compare the data as we developed the simulations. Disturbingly, our initial data was not to our liking.

We started using the high speed camera, which is something that we did not use before during our initial familiarization and we saw some disturbing images. At low frequencies ($< 12$hz), we saw that the vertical oscillation were being severally dampened. Instantly we thought that this would be a major problem since the easier patterns of maximums and minimums that we could see would come form the lowest frequencies. The higher frequencies produced greater number of modes and their smaller amplitudes were very hard to make out. Because our simulations had not yet been finalized, we could not tell how big of an effect this would have.

As we continued to what the video, we notice that the CD container was teeter tottering from side to side. We found that the table was not levelled properly. We re-levelled the table and found that the mounting screw that protruded through the plexiglass was not perfectly perpendicular to the surface so that there was a small angle of error in the vertical axis.

Another problem that we noticed was the confirmation of the sinusoidal peaks being flattened at the top. We had noticed this before in our initial familiarization with the equipment, but didn't have much appreciation for it then. As we adjusted the frequency, we noticed that it went away when the frequency was increase past 2hz. Form this we realized that we could use frequencies lower than the said value. We chose 2.5hz to be our lowest range of frequencies to run.

After some frustration, we learned of yet another problem and that had to do with the wave generator. This is the electronic piece of equipment that produces the sin wave that goes to the vertical oscillator. We found that there was approximately a $+/-0.03$hz fluctuation in the frequency being generated. We didn't know whether or not this would be a problem until we observed that if the frequency was adjusted just a hundredth of a hz, the pattern would change drastically as well (and the knob was very touchy).

If our project was to prove successful, we would have to do something about the noise.

# 6 Fixing the Noise

One must take into account the fact that our equipment is not the best and it is going to produce some deviation from our computer models. Therefore, we have to reduce the errors in the system to as much of a minimum as possible.

For the dampening we decided to reduce the volume of water inside because we felt that reducing the mass would reduce the dampening. The dampening was reduced but was still significantly present and we did not want reduce the fluid level below 1cm, for then the minimums would bottom out on the bottom surface of the container. At this moment, the team replaced the now needlessly tall container for a 2.5cm height container with a radius of '7cm and a water height of 0.7cm. The volume did not change much so the oscillations were still present. The replacement however, did eliminate the teeter tottering effect. Since we could not due anymore to reduce/eliminate the dampening, we just choose to go with what we had and see if it matched up with our data.

The flattened peaks were easily avoided by using driving frequencies $> 2.5$hz.

For the frequency oscillations of $+/-0.03$hz, we had no choice but to use the equipment for a lack of no other. We would have to wait and see if it would have an effect.

# 7 Data Collecting and Comparison

When the first series of data collecting took place, our frequency range was 2.5hz, 5hz, 10hz, 15hz, 20hz, 25hz, 30hz, 35hz, 40hz, 60hz and 100hz. We took high speed video footage of the runs so that we could play them back in slow motion. From that we would analyze the modes and then compare them with the MAPLE computer simulations. This became a minor setback.

By now we had our computer simulations running and realized that we should have run the computer simulations first and then re-ran them in the laboratory. The reason is that we learned that in the computer simulations, our input values are the Bessel modes m and n and from that we can get our driving frequency out. On the contrary, in the laboratory, we put our input values of our driving frequency into the wave generator to see what pattern, or modes m and n are produced. Since we could not put m and n into the laboratory and we could put the driving frequency into MAPLE, we had to take the output frequency from MAPLE and put it into the laboratory equipment. This made the initial pile of data practically worthless since MAPLE was giving us a different range of frequencies.

We therefore had to run the simulations first and then take the frequency that was generated and put it into the wave generator to see if the surface pattern was produced. The frequencies that MAPLE was producing were alarming in that the values were so close to each other (by only a couple hundredths of a hz). To get the wave generator that precise was impossible yet the results were not completely a failer. It turned out that if we tweaked the knob very meticulously up and down in range, we were able to see some of the patterns. However, the reader must appreciate how difficult it is to make out the patterns with the video. Frame by frame had to be moved until we could see some of the maximums and minimums and not all of the patterns were replicated. For modes that could be seen in MAPLE, the frequencies were all below 10hz. Remember that in this range (2.5hz to 10hz) the waves are dampened and there is still a lot of noise. Only above 12hz does the dampening begin to disappear but the noise is still there. Also, while watching the video we could see many smaller sporadic waves that appeared to be from the noise of the equipment. Some of the comparisons are shown in

the following figures.

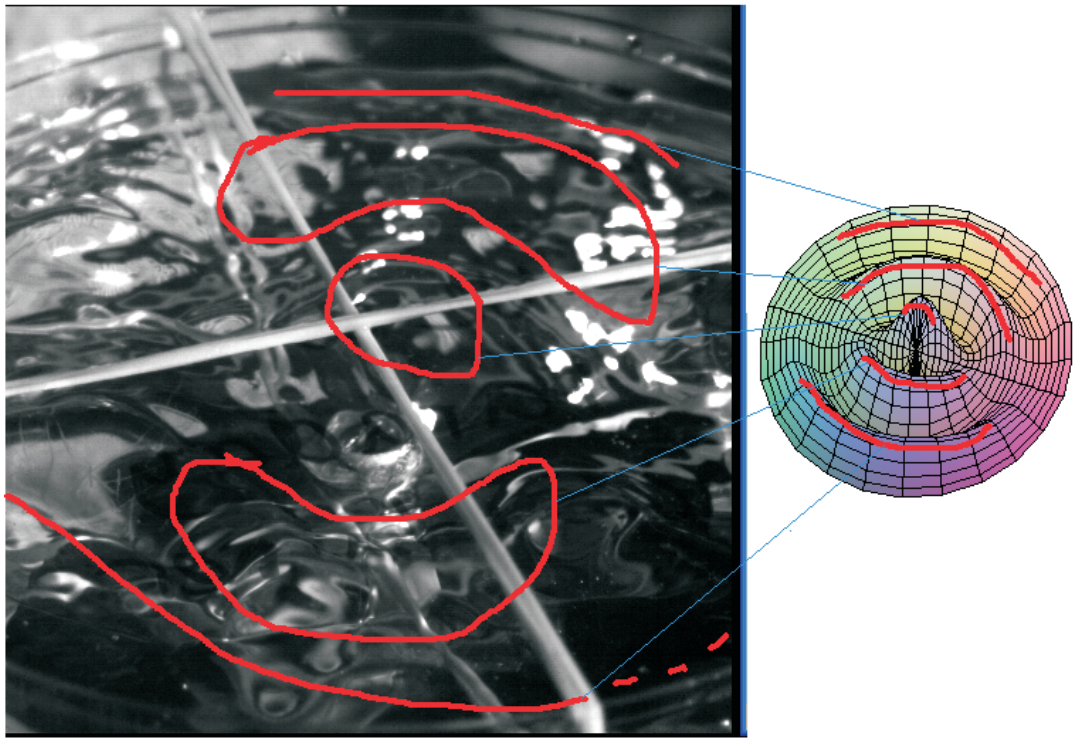

Figure 4: Comparison for m=1;n=4. Peaks outlined in red and crossed referenced in blue.

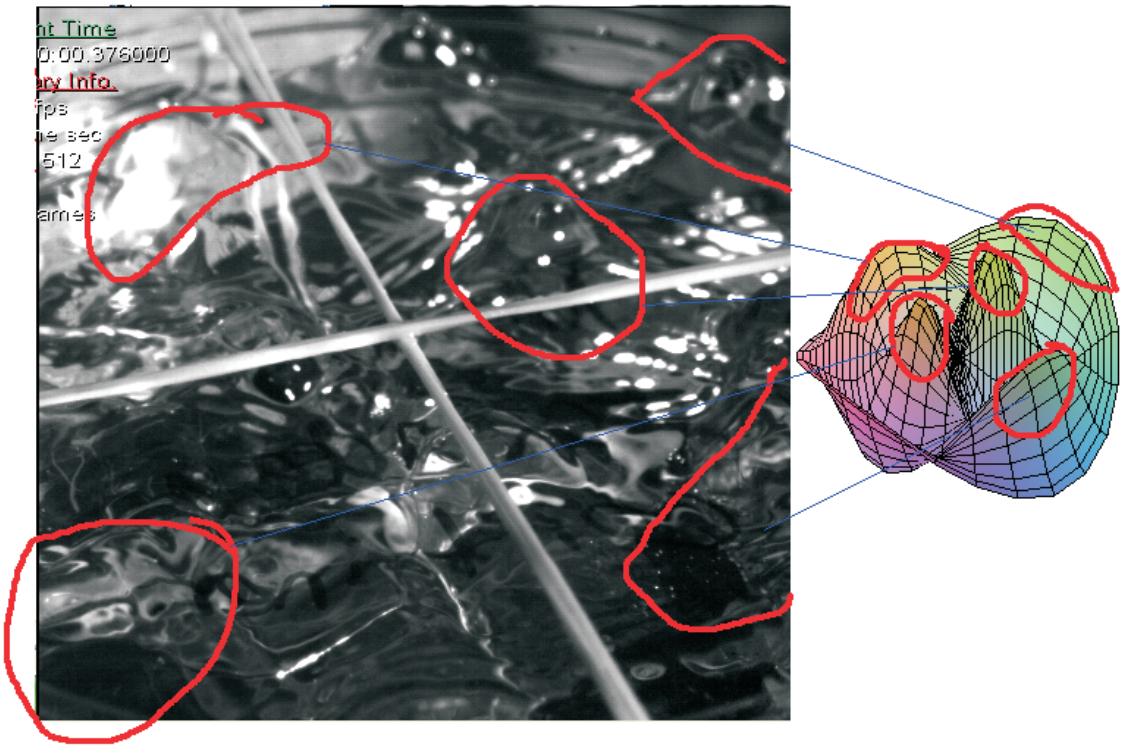


Figure 5: Comparison for m=2;n=3. Peaks outlined in red and crossed referenced in blue.

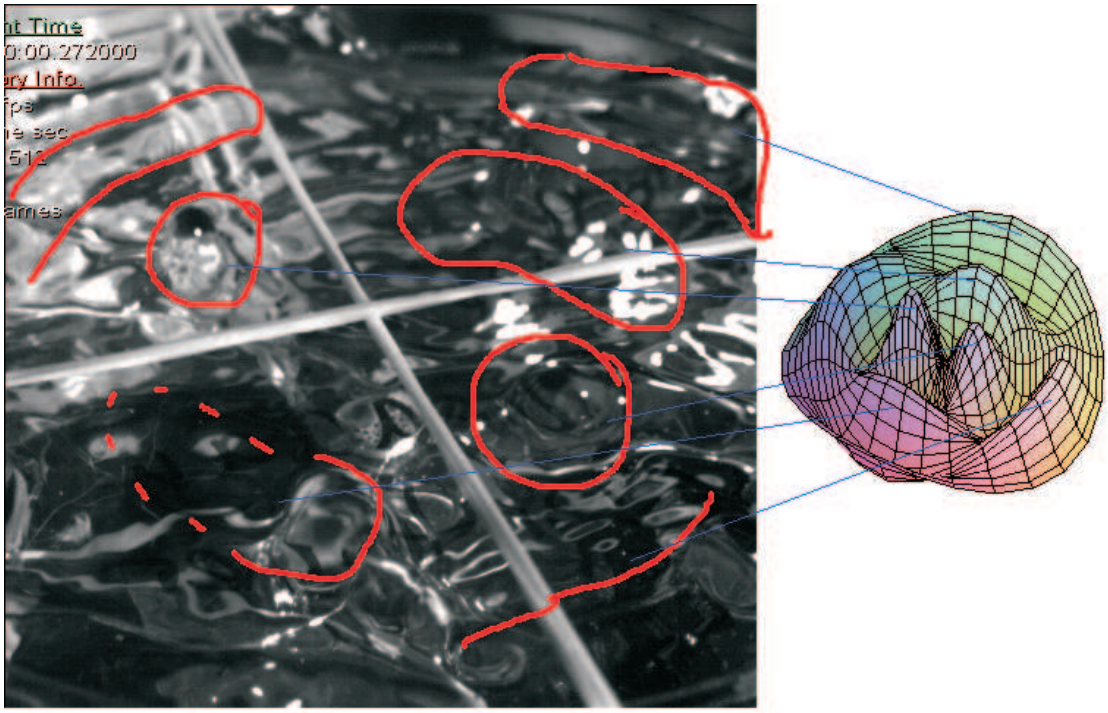


Figure 6: Comparison for m=2;n=4. Peaks outlined in red and crossed referenced in blue.

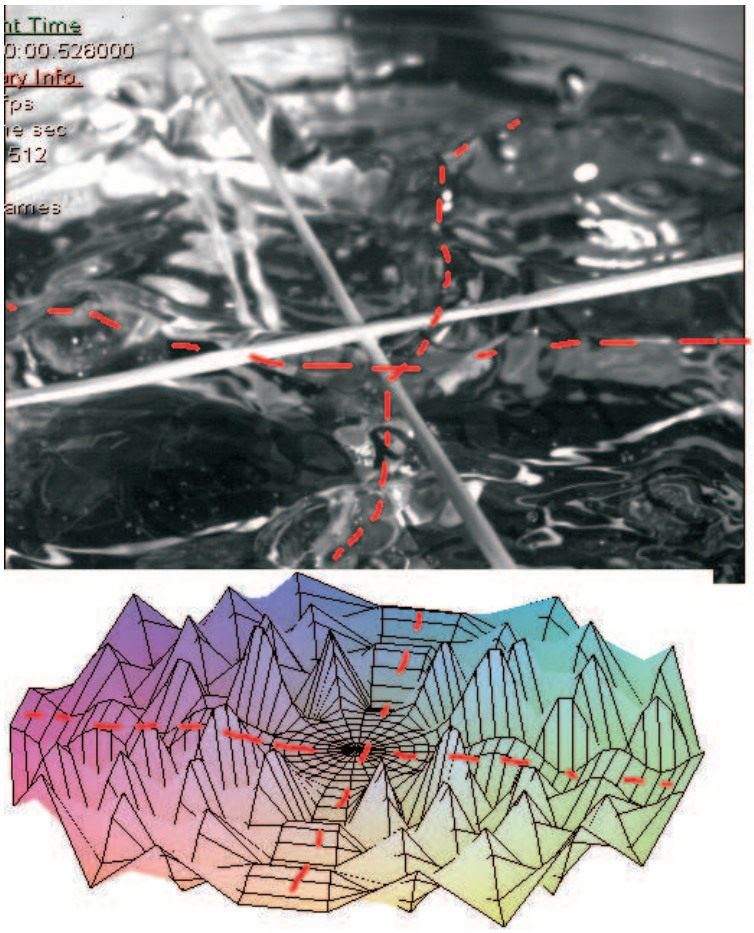


Figure 7: Comparison for m=10;n=5. This time it was harder to see the peaks so instead the easier pattern to see was when the maximums went to the minimums. Notice they appear to be the inverse of each other.

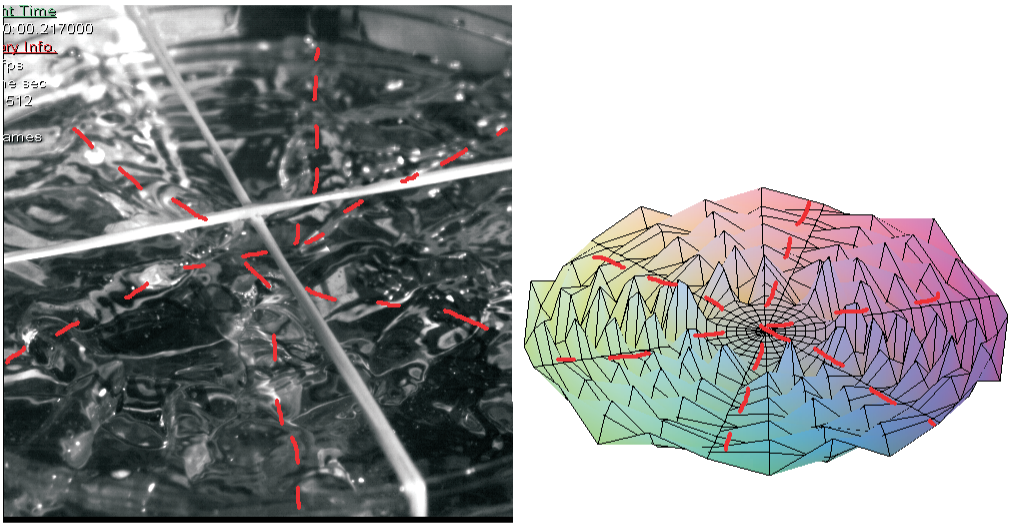

Figure 8: Comparison for m=15;n=10. Another inverse pattern.

# 8 Conclusion

In this project our team analyzed the surface patterns on the surface of a fluid in a cylindrical container. We made computer simulations and tested them against data from a real laboratory experiment. We learned that to some extent we were successful in that some off our simulations worked and we replicated them in the lab. However, we feel that the equipment might be inadequate for this experiment. Either the vertical oscillator or the source generator was not strong enough to handle the weight of the water and container. The oscillating table probably needs more support as well. Because the equipment seems to be plagued with noise, we do not feel that there was a problem with our computer simulation. However, even if there was, we would not be able to test it against the laboratory equipment since there is just too much noise in the equipment.

# 9 Future Projections

If a more powerful combination of a vertical oscillator and signal generator were available, we would recommend that further experiments be conducted. There are also other shapes of containers that could be tested, such as a rectangular shape or an oval shape. Also the list of fluids that could be used is also quite extensive. There could be non-Newtonian fluids that are used as well as objects that are inserted into/onto the fluid in order to dampen the surface waves. In addition, our experiment could be duplicated for validity and reliability.

# 10 Acknowledgements

The Math 512 Faraday team would like to thank Dr. Rossi for his guidance and assistance in this project, our teaching assistant Lauren Rossi for providing us with some helpful hints at succeeding in the class (esp. with the milestone), Geoff Oxberry who helped us clean up and fix our math so that we could move on and to Brandon Farmer who has cleared our names from the BIBTEX hit list forever.

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